MIL-S-82517A(OS)
23 April 1974
Superseding
MIL-S-82517(OS)
17 November 1967

MILITARY SPECIFICATION

SWITCH, HYDROSTATIC

MARK 58 MODS 0 AND 1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes the requirements of one type of Hydrostatic Switch, Mark 58 Mods 0 and 1. (See 6.1 for intended use and description.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATION

Federal

PPP-B-621	Boxes, Wood, Nailed and Lock Corner

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible

FSC 1135

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STANDARDS

Federal

FED-STD-102	Preservation, Packaging, and Packing Levels

* Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment
MIL-STD-810	Environmental Test Methods

DRAWINGS

* Naval Ordnance Systems Command (Code Ident 10001)

LD 490174	Switch, Hydrostatic, Mark 58 Mod 0 and all drawings listed thereon
DL 1984367	Dynamic Tester, Hydrostatic Switch; Assembly
DL 1984372	Switch, Hydrostatic, Mark 58 Mod 1 and all drawings listed thereon

PUBLICATIONS

* Naval Ordnance Systems Command (Code Ident 10001)

OD 41570	Operation and Maintenance of the Dynamic Tester for Hydrostatic Switch Mk 58 Mods 0 and 1

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

* 3.1 General requirements. Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in the manufacture of the switch shall be in accordance with the requirements specified herein and with the applicable documents listed in NAVORD LD 490174 or DL 1984372.

3.2 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the switch shall be manufactured using the methods and procedures proposed for production. The sample shall be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.3 Performance and product characteristics. The switch shall meet the following performance and product characteristics:

3.3.1 Circuit requirements

3.3.1.1 Normally closed circuit. The normally closed circuit shall have a resistance of not greater than 0.10 ohm.

3.3.1.2 Insulation resistance. After being conditioned at a temperature of 75 plus or minus 5 degrees Fahrenheit (F) and a relative humidity of 75 plus or minus 5 percent for a period of not less than 24 hours, the insulation resistance shall be not less than 100 megohms. When the insulation resistance is measured during or subsequent to the environmental tests as specified in 4.5, there shall be no temperature or humidity conditioning other than that required by the individual environmental test and the insulation resistance shall then be 20 megohms minimum. The test potential shall be 500 plus or minus 50 volts dc. The insulation resistance shall be measured between the following mutually insulated points:

(a) Case

(b) One lead of the normally open circuit

(c) Other lead of the normally open circuit

(d) Either lead of the normally closed circuit

* 3.3.2 Functioning pressure

* 3.3.2.1 Static test, normally closed circuit. The switch shall not open the normally closed circuit when subjected to a pressure of 10 psig or less. After the circuit is opened by the application of increased pressure, the switch shall close the circuit when the pressure returns to atmospheric.

* 3.3.3 Functioning Times

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* 3.3.3.1 Functioning times before environment. The switch shall, before experiencing any environmental tests, open the normally closed circuit within 1.23 seconds maximum after pressurization. The switch shall close the normally open circuit 1.71 plus or minus 0.20 seconds after pressurization.

* 3.3.3.2 Functioning times after environment. The switch shall, after experiencing any environmental tests, open the normally closed circuit within 1.23 seconds maximum after pressurization. The switch shall close the normally open circuit 1.71 plus or minus 0.50 seconds after pressurization.

3.3.4 Circuit requirements, under pressure

3.3.4.1 Normally open circuit. The normally open circuit shall have a resistance of not greater than 0.10 ohm when the switch is subjected to a pressure of 75 psig.

3.3.4.2 Insulation resistance. After being conditioned at a temperature of 75 plus or minus 5 degrees F and a relative humidity of 75 plus or minus 5 percent for a period of not less than 24 hours, and when subjected to a pressure of 75 psig, the switch shall have an insulation resistance of not less than 100 megohms. When the insulation resistance is measured during or subsequent to the environmental tests as specified in 4.5, there shall be no temperature or humidity conditioning other than that required by the individual environmental test, and the insulation resistance shall then be 20 megohms minimum. The test potential shall be 500 plus or minus 50 volts dc. The insulation resistance shall be measured between the following mutually insulated points:

(a) Case

(b) One lead of the normally closed circuit

(c) The other lead of the normally closed circuit

(d) One lead of the normally open circuit

3.4 Environmental requirements. Unless otherwise specified, the switch shall be capable of conforming to the performance and product characteristics of this specification after being exposed to the environments and all conditions specified herein.

3.4.1 Current capacity. The switch circuits shall be capable of carrying a current of 3 amperes continuously.

3.4.2 Temperature, storage condition. The switch shall withstand temperatures from minus 65 to plus 160 degrees F.

* 3.4.3 Humidity. The switch shall withstand relative humidity up to 95 percent as specified in MIL-STD-810, Method 507 Procedure I, on the exposed area as shown in Figure 1.

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* 3.4.4 Fungus. The switch shall withstand fungus conditions as specified in MIL-STD, Method 508, Procedure I, on the exposed area as shown in Figure 1.

* 3.4.5 Dust. The switch shall withstand dust conditions as specified in MIL-STD-810, Method 510, Procedure I, on the exposed area as shown in figure 1.

* 3.4.6 Salt fog. The switch shall withstand salt fog conditions as specified in MIL-STD-810, Method 509 Procedure I. on the exposed area as shown in Figure 1.

3.4.7 Vibration, standby condition. The switch shall withstand vibration as specified for Type I equipment in MIL-STD-167 up to 25 Hertz (Hz).

3.4.8 Shock, standby condition. The switch shall withstand shocks of 40g (g=unit of gravitational acceleration) peak magnitude. The shock shall have a pulse shape approximating a one-half sine wave having a duration or 5 milliseconds.

3.4.9 Acceleration. The switch shall withstand an acceleration of 25g for 4 seconds applied in the direction shown in Figure 2.

3.4.10 Vibration, operating condition. The switch shall withstand vibration through a frequency range of 20 to 2000 Hz at 0.10-inch double amplitude or 10g, whichever is less.

3.4.11 Shock, operating condition

3.4.11.1 Major. The switch shall withstand a shock of 2400g peak magnitude in the direction shown in Figure 2. The shock shall have a pulse shape approximating a one-half sine wave having a duration of 1 millisecond.

3.4.11.2 Minor. The switch shall withstand two shocks of 1200g peak magnitude 90 degrees apart and perpendicular to the major shock axis. The shock shall have a pulse shape approximating a one-half sine wave having a duration of 1 millisecond.

3.4.12 Temperature, operating condition. The switch shall meet the performance and product characteristics while at temperatures from 35 to 120 degrees F.

3.4.13 Pressure. The switch shall withstand a hydrostatic pressure of 750 psig on the exposed area shown in Figure 1 without leaking or shorting the normally open circuit, closed by the pressure, to the case.

3.5 Workmanship. The workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with the requirements of this specification. All parts used in the assembly shall be thoroughly clean and free of excess materials, chips, and loose or splattered

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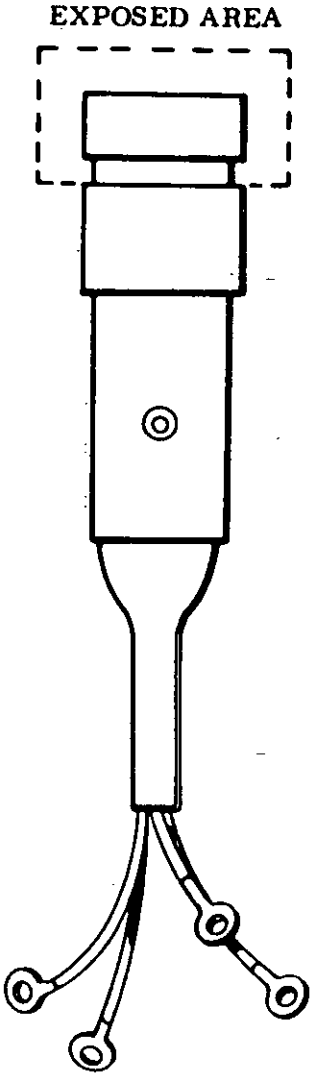


Figure 1. Environmental Test

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foreign materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the switch shall be classified as follows:

(a) Preproduction and periodic production inspection (See 4.5.)

(b) Quality conformance inspection (See 4.6.)

4.3 Sampling. (See 6.2.1.)

4.3.1 Lot size. The term "lot" shall mean "inspection lot", that is, a collection of product submitted by a supplier for inspection. Unless otherwise specified in the contract or order, the number of product in each lot shall be 300. Lot formation shall be in accordance with MIL-STD-105.

4.3.2 Preproduction sample. A preproduction sample of 15 switches, manufactured in accordance with 3.2, shall be subjected to the preproduction tests detailed in 4.5. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the switch by the supplier, prior to approval of the preproduction sample, shall be at the supplier's risk.

4.3.3 Periodic production sample. From each succeeding lot of 300 or less, a random sample of eight switches shall be selected. These sample units shall be subjected to the tests of 4.5. The switches shall be considered satisfactory if they meet all specification and pertinent drawing requirements and withstand the tests specified in 4.5 (See Table I). When a defective unit is found in any lot, acceptance of all units in the lot shall be withheld and the contractor shall:

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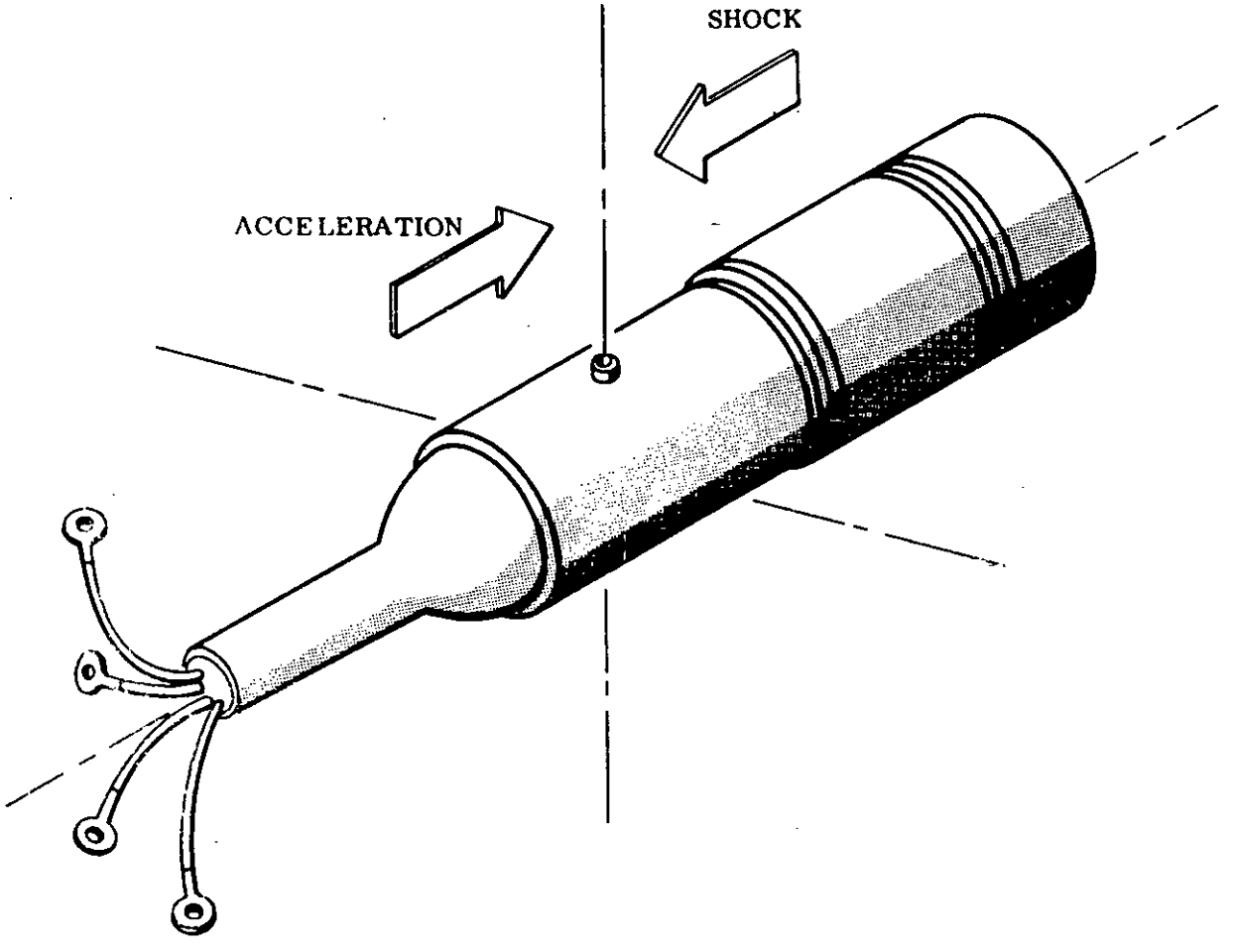


Figure 2. Environmental Testing Axes

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(a) Determine the cause of the defect.

(b) Take the necessary action, with the concurrence of the procuring activity, to correct or eliminate the defect from all units in the lot and, if so directed by the procuring activity, from those units previously shipped.

(c) Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

* 4.3.4 Quality conformance inspection. Each switch shall be inspected as specified in 4.6.1 to determine its acceptibility.

4.4 Test equipment and test conditions

4.4.1 Quality conformance inspection equipment. The following items of test equipment are required to perform the quality conformance inspection tests:

	Equipment	Capability
(a)	Low resistance measuring device	0.00 to 0.10 ohm; full-scale accuracy of 5 percent.
(b)	Conditioning chamber	Maintain temperature at 75 plus or minus 5 degrees F and a relative humidity of 75 plus or minus 5 percent.
(c)	Megohmmeter	0 to 100 megohms; accuracy 10 percent at 20 and 100 megohms with 500 plus or minus 50 volts dc potential.
* (d)	Dynamic Tester, Hydrostatic Switch	NAVORD DL 1984367

4.4.2 Preproduction and periodic production test equipment. The following items of test equipment, in addition to selected items of quality conformance inspection equipment and pieces of equipment recommended by applicable specifications or standards referenced within test procedure paragraphs are required to perform the preproduction and periodic production tests:

	Equipment	Capability
(a)	Voltmeter	0.00 to 0.50 volt dc; accuracy ±3 percent

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Table I. Preproduction and Periodic Production Test Sequence

Test	Paragraph Number	Test Quantities	
		Preproduction	Periodic
Quality Conformance Inspection	4.6*	15	8
Current Capacity	4.5.1.1	15	8
Temperature, Storage Condition	4.5.1.2	15	8
Humidity	4.5.1.3	15	-
Fungus	4.5.1.4	15	-
Quality Conformance Inspection	4.6**	15	8
Dust	4.5.1.5	15	-
Salt Fog	4.5.1.6	15	8
Vibration, Standby Condition	4.5.1.7	15	8
Shock, Standby Condition	4.5.1.8	15	8
Quality Conformance Inspection	4.6**	15	8
Acceleration	4.5.1.9	15	8
Vibration, Operating Condition	4.5.1.10	15	8
Shock, Operating Condition	4.5.1.11	15	8
Temperature, Operating Condition Low Extreme Quality Conformance Inspection	4.5.1.12 4.5.1.12.1 4.6**	15	8
High Extreme Quality Conformance Inspection	4.5.1.12.2 4.6**	15	8
Pressure	4.5.1.13	15	8

*Does not include the test of 4.6.1.3.2
**Does not include the test of 4.6.1.3.1

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Equipment	Capability
(b) DC power supply, current regulated	Regulate direct current at 3.00 plus 0.00 minus 0.30 amperes through test item load (approximately 0.1 ohm maximum).
(c) Temperature chamber	Maintain temperature at: Minus 65 plus 5 minus 0 degrees F Plus 35 plus 5 minus 0 degrees F Plus 120 plus 0 minus 5 degrees F Plus 160 plus 0 minus 5 degrees F
(d) Vibration testing machine	Control frequency from 20 Hz to 2000 Hz at 0.10-inch double amplitude or 10g acceleration, whichever is less; maintain direction of vibration along specified axes.
(e) Centrifuge	0 to 27.5g; maintain acceleration at 25 plus 2.5 minus 0g.
(f) Shock equipment, operating	Apply shocks of up to 2400g plus or minus 10 percent; a one-half sine wave pulse shape of 1 plus or minus 0.25 millisecond time duration; maintain direction along specified axes.

4.4.3 Test conditions. Unless otherwise specified, the switch shall be subjected to the preproduction and periodic production tests and quality conformance inspection tests under the following conditions.

(a) Temperature	Room ambient (65 to 95 degrees F)
(b) Altitude	Normal ground atmospheric pressure
(c) Humidity (Relative)	Room ambient up to 95 percent

4.5 Preproduction and periodic production inspections. The preproduction and periodic production samples shall be subjected to the tests specified in 4.5.1 after satisfactorily passing the quality conformance inspections of 4.6, except 4.6.1.3.2.

4.5.1 Test procedure. Subject the switch to the test sequence listed in Table I. Unless otherwise specified in the individual test paragraph, the quality conformance inspection as specified in Table I shall be performed subsequent to environmental tests under test conditions specified in 4.4.3. Failure of the switch to pass any one of these tests shall constitute a defective unit.

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4.5.1.1 Current capacity test.

4.5.1.1.1 Normally closed circuit. Connect the voltmeter, specified in 4.4.2(a), across the leads of the normally closed circuit and apply a direct current of 3.00 plus 0.00 minus 0.30 amperes for a duration of 90 seconds minimum, using the power supply specified in 4.4.2(b). The voltage drop shall be not greater than 0.30 volt.

* 4.5.1.1.2 Normally open circuit. Using the tester specified in 4.4.1(d), subject the switch to a pressure of 75 psig minimum in accordance with OD 41570. Using the power supply specified in 4.4.2(b), apply a direct current of 3.00 plus 0.00 minus 0.30 amperes for a duration of 90 seconds minimum to the leads of the normally open circuit. The voltage drop shall not exceed 0.30 volt across the leads of the normally open circuit, as measured by the voltmeter specified in 4.4.2(a).

4.5.1.2 Temperature, storage condition. The switch shall be subjected to a temperature of minus 65 plus 5 minus 0 degrees F for not less than 48 hours. The temperature shall be raised to plus 160 plus 0 minus 5 degrees F in not less than 2 hours and maintained at this temperature for not less than 48 hours. The unit shall then be subjected to room temperature for not less than 4 hours prior to performing subsequent tests.

* 4.5.1.3 Humidity. Unless otherwise specified in the contract or order, this test shall be performed on preproduction sample units only. Switches shall be tested in accordance with Method 507, procedure I of MIL-STD-810, on the exposed area shown in figure 1.

* 4.5.1.4 Fungus. Unless otherwise specified in the contract or order, this test shall be performed on preproduction sample units only. Switches shall be tested in accordance with Method 508, procedure I of MIL-STD-810, on the exposed area as shown in Figure 1.

* 4.5.1.5 Dust. Unless otherwise specified in the contract or order this test shall be performed on preproduction sample units only. Switches shall be tested in accordance with Method 510, procedure I of MIL-STD-810, on the exposed area as shown in figure 1.

* 4.5.1.6 Salt fog. Switches shall be tested in accordance with Method 509, procedure I of MIL-STD-810, on the exposed area as shown in Figure 1.

4.5.1.7 Vibration, standby condition. The switch shall be subjected to the Vibration tests for Type I equipment as specified in MIL-STD-167 up to 25 Hz in each of the three axes shown in Figure 2. Mount the switch as shown in Figure 3.

NOTE: Instrumentation used for the shock test shall not employ a filter with a cut-off frequency below 10 kHz.

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4.5.1.8 Shock, standby condition. Subject the switch to shocks of 40 plus or minus 4g peak magnitude with a pulse shape approximating a one-half sine wave, having a duration of 5 plus 3 minus 0 milliseconds. Shocks shall be applied 3 times in each direction along each of 3 mutually perpendicular axes as shown in Figure 2. Mount the switch as shown in Figure 3.

4.5.1.9 Acceleration. The switch, mounted as shown in Figure 3, shall be subjected to acceleration of 25 plus 2.5 minus 0g in the direction shown in Figure 2 for not less than 4 seconds.

4.5.1.10 Vibration, operating condition. The switch, mounted as shown in Figure 3, shall be subjected to vibration from 20 to 2000 Hz at 0.10-inch double amplitude or 10g, whichever is less, along each of 3 mutually perpendicular axes. The frequency shall be increased uniformly from 20 to 2000 Hz. The frequency shall be traversed in not less than 2.5 minutes, once in each axis.

4.5.1.11 Shock, operating condition. The switch shall be mounted as shown in Figure 3, and subjected to the following tests.

4.5.1.11.1 Major. Subject the switch to a shock of 2400g plus or minus 240g peak magnitude. The shock shall have a pulse shape approximating a one-half sine wave having a duration of 1 plus or minus 0.25 millisecond. Apply the shock in the direction shown in Figure 2.

4.5.1.11.2 Minor. Subject the switch to two shocks of 1200 plus or minus 120g peak magnitude. The shocks shall have a pulse shape approximating a one-half sine wave having a duration of 1 plus or minus 0.25 millisecond. Apply the shocks in two directions 90 degrees apart and perpendicular to the major shock axis.

4.5.1.12 Temperature, operating condition

4.5.1.12.1 Low extreme. The switch shall be subjected to a temperature of 35 plus 5 minus 0 degrees F for not less than 2 hours, and while at that temperature shall pass the quality conformance inspection tests.

4.5.1.12.2 High extreme. The switch shall be subjected to a temperature of 120 plus 0 minus 5 degrees F for not less than 2 hours, and while at that temperature shall pass the quality conformance inspection tests.

4.5.1.13 Pressure. The switch shall be subjected to a hydrostatic pressure of 750 plus or minus 25 psig, applied to the exposed area as shown in Figure 1 for a duration of not greater than 30 seconds, during which the switch shall not leak and the normally open circuit, closed by pressure, shall not short to the case.

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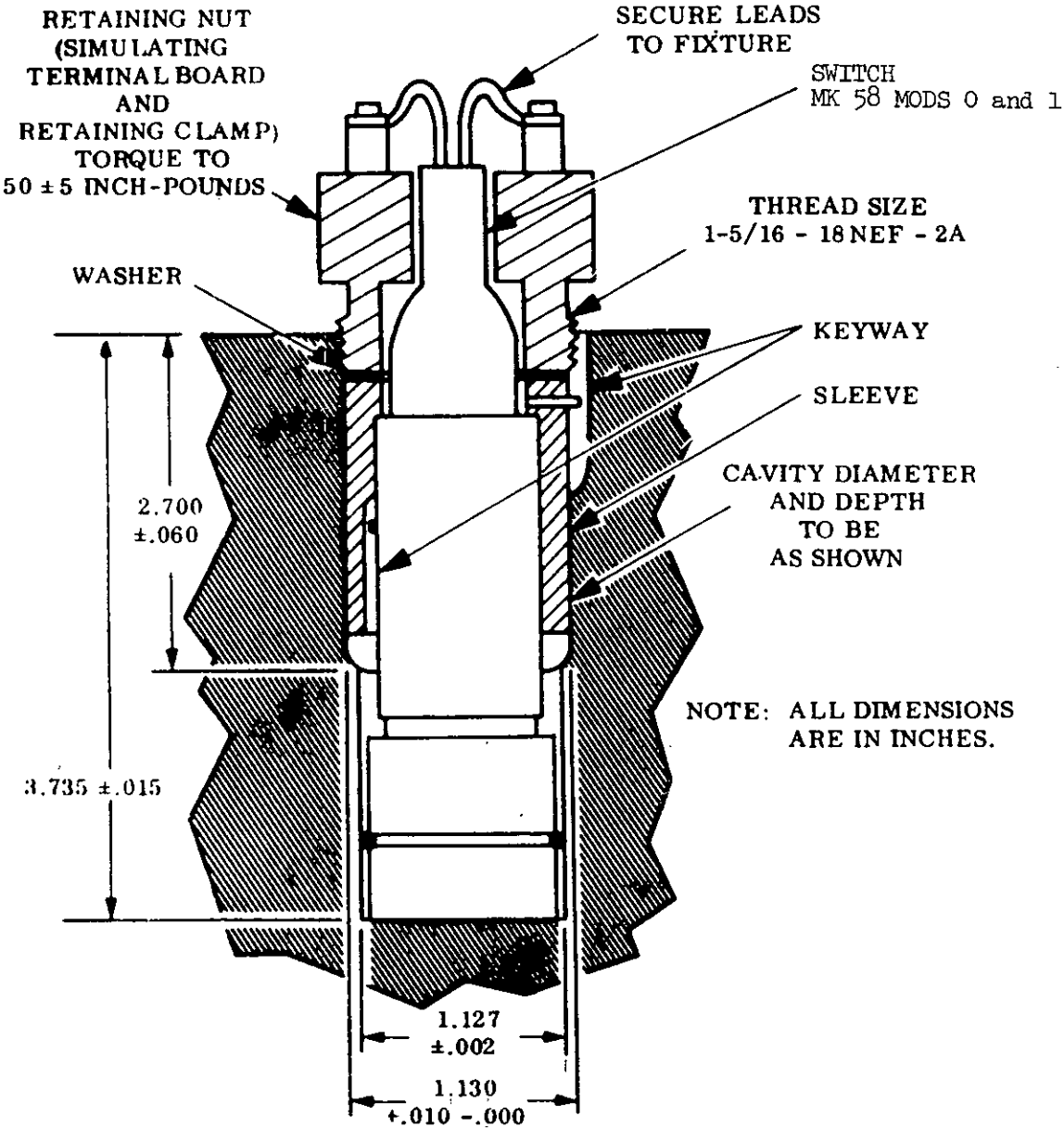


Figure 3. Suggested Vibration, Acceleration, and Shock Test Fixture

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* 4.6 Quality conformance inspection. Each switch, using the plans, procedures, and sequence set forth in Table II of this specification, shall be subjected to the tests and examinations as described in 4.6.1.

4.6.1 Test Procedure. The switch shall be subjected to the tests and examinations as outlined herein. Unless otherwise specified, inability of the switch to pass the tests and examinations shall constitute a failure.

4.6.1.1 Circuit requirements.

4.6.1.1.1 Normally closed circuit. Using the low resistance measuring device specified in 4.4.1(a), measure the resistance of the normally closed circuit. The resistance shall be not greater than 0.10 ohm.

* Table II. Quality Conformance Tests and Examinations and AQL.

Test	Sampling Plan	Test Paragraph
Circuit requirements		4.6.1.1
Normally closed circuit	**	4.6.1.1.1
Insulation resistance	**	4.6.1.1.2
Functioning pressure		4.6.1.2
Static test, normally closed circuit	**	4.6.1.2.1
Functioning times		4.6.1.3
Functioning times before environment	**	4.6.1.3.1
Functioning times after environment	*	4.6.1.3.2
Circuit requirements under pressure		4.6.1.4
Normally open circuit	**	4.6.1.4.1
Insulation resistance	**	4.6.1.4.2
Visual examination	**	4.6.1.5
Preservation, packaging, packing, and marking	**	4.7

**One-hundred percent inspection; defect is cause for rejection of individual unit.

* One-hundred percent inspection; defect is cause for rejection of the lot. Applies only to preproduction and periodic test samples.

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4.6.1.1.2 Insulation resistance. Place the switch in a conditioning chamber and maintain the temperature at 75 plus or minus 5 degrees F and the relative humidity at 75 plus or minus 5 percent for a period of not less than 24 hours. Remove the switch from the chamber and within 10 minutes measure the insulation resistance between the mutually insulated points listed below. The applied potential shall be 500 plus or minus 50 volts dc. The insulation resistance shall be not less than 100 megohms. When this test is performed during or subsequent to the environmental tests specified in 4.5.1, there shall be no temperature or humidity conditioning other than that required by the individual environmental test; the insulation resistance shall then be 20 megohms minimum. The mutually insulated points are:

(a) Case

(b) One lead of the normally open circuit

(c) Other lead of the normally open circuit

(d) Either lead of the normally closed circuit

4.6.1.2 Functioning pressure

* 4.6.1.2.1 Static test, normally closed circuit. Using the tester specified in 4.4.1(d), subject the switch to a pressure of 10 psig minimum in accordance with OD 41570. The normally closed circuit shall not open. Subject the switch to a pressure sufficient to open the circuit but not greater than 50 psig, and then lower the pressure to atmospheric. The normally closed circuit shall close.

** 4.6.1.3 Functioning times. Using the tester specified in 4.4.1(d), subject the switch to the dynamic test in accordance with OD 41570. After the switch is fully pressurized the normally open circuit shall not open. When the pressure is lowered to atmospheric the normally closed circuit shall close.

** 4.6.1.3.1 Functioning times before environment. The switch, shall, before experiencing any of the environmental tests specified in 4.5.1, open the normally closed circuit within 1.23 seconds maximum after pressurization. The switch shall close the normally open circuit 1.71 plus or minus 0.20 seconds after pressurization.

* 4.6.1.3.2 Functioning times after environment. The switch shall, after experiencing any of the environmental tests specified in 4.5.1, open the normally closed circuit within 1.23 seconds maximum after pressurization. The switch shall close the normally open circuit 1.71 plus or minus 0.50 seconds after pressurization.

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4.6.1.4 Circuit requirements, under pressure

4.6.1.4.1 Normally open circuit. Using the tester specified in 4.4.1(d), subject the switch to a pressure of 75 psig minimum. Using the low resistance measuring device specified in 4.4.1(a), measure the resistance of the normally open circuit. The resistance shall be not greater than 0.10 ohm.

4.6.1.4.2 Insulation resistance. Place the switch in a conditioning chamber and maintain the temperature at 75 plus or minus 5 degrees F and the relative humidity at 75 plus or minus 5 percent for a period of not less than 24 hours. Remove the switch from the chamber and within ten minutes, using the megohmmeter and the tester specified in 4.4.1(c) and 4.4.1(d) respectively, subject the switch to a pressure of 75 psig minimum and measure the insulation resistance between the mutually insulated points listed below. The insulation resistance shall be not less than 100 megohms, when the applied potential is 500 plus or minus 50 volts dc. When this test is performed subsequent to the tests specified in 4.5.1, the insulation resistance shall then be 20 megohms minimum. The mutually insulated points are:

(a) Case

(b) One lead of the normally closed circuit

(c) The other lead of the normally closed circuit

(d) One lead of the normally open circuit

4.6.1.5 Visual examination. Visual examination shall be made on each switch submitted for delivery to the procuring activity. The switch shall be examined for conformance to the requirements of good workmanship in 3.5. Nonconformance is cause for rejection of individual unit.

4.7 Preservation, packaging, packing, and marking. The switch shall be examined for conformance to the prescribed preservation, packaging, packing, and marking requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified. (See 6.2(c) and 6.4.)

5.1.1 Level A

5.1.1.1 Cleaning. The switch shall be cleaned in accordance with MIL-P-116, Method C-1.

5.1.1.2 Drying. The switch shall be dried in accordance with MIL-P-116, Method D-1.

5.1.1.3 Unit packaging. The switch shall be individually packaged in accordance with MIL-P-116, Method IC-1.

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5.1.2 Level C. Cleaning, drying, preservation, and packaging shall be in accordance with the supplier's commercial practice, when such practices have been approved by the procuring activity.

5.2 Packing. Packing shall be Level A or C, as specified. (See 6.2(d) and 6.4.)

5.2.1 Level A. Unless otherwise specified in the contract, order or requisition, the switch shall be packed in exterior-type shipping container, PPP-B-621, of uniform size and shape, with the minimum cube and tare consistent with the protection required. The gross weight of each pack shall be limited to approximately 100 pounds. Containers shall be provided with a case liner, MIL-L-10547, and shall be closed and strapped in accordance with PPP-B-621.

5.2.2 Level C. The switch, packaged as specified in 5.1.2, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use and description

6.1.1 Intended use. The switch is intended for use in a depth charge to sense water depth by pressure.

6.1.2 Description. The switch is a pressure-operated device. The unit consists essentially of a diaphragm, a plunger, a spring and two sets of electrical contacts. When subjected to a predetermined pressure, the force applied to the face of the diaphragm causes the plunger to move, depressing the spring. This motion causes a switch bar to open the normally closed set of contacts and to close the normally open set of contacts.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, Number, and date of this specification.

(b) Mod 0 or Mod 1 required.

(c) Required level of preservation and packaging.

(d) Required level of packing.

(e) Any exceptions to this specification.

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6.2.1 Additional samples. Additional samples required because of failure of any sample(s) to pass the prescribed inspections shall be provided and inspected at the expense of the supplier.

6.3 Definitions. For the purpose of this specification the following definitions have been established.

6.3.1 Unit condition

6.3.1.1 Operating. Operating is the condition of the switch when performing its required function.

6.3.1.2 Standby. Standby is the condition of the switch when the switch is aboard ship and is:

(a) Stored, handled or transported

(b) Assembled with the end item and installed on the launching mechanism prior to operation

6.4 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as specified in FED-STD-102.

6.5 Marginal identification. The margins of this specifcation are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Project No. 1135-N011

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-S-82517A(OS), Switch, Hydrostatic MARK 58 MODS 0 AND 1

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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